

# **Systems and Cognitive Neuroscience (NEU 502A / PSY 502A / MOL 502A)**

Spring 2025

**Professor:** Jonathan Cohen (jdc@princeton.edu)

**AI:** Alex Ku (alexku@princeton.edu)

## **Course Description**

A survey of experimental & theoretical approaches to understanding how cognition arises in the brain. This complements NEU 501A, focusing on the mechanisms responsible for perception, attention, decision making, memory, cognitive & motor control, and planning, with emphasis on the representations involved & their transformations in the service of cognitive function. Source material will span neuroscience, cognitive science, & work on artificial systems. Relevance to neurodegenerative and neuropsychiatric disorders will also be discussed. This is the 2nd term of a double-credit core lecture course required of all Neuroscience Ph.D. students.

## **Course Structure**

*Lectures:* M/Th 2-4:30

*Format:* the two classes each week will be divided into three parts: “orienting” lectures, a “deep dive” by a visiting lecturer, and a student paper presentation. There will be two orienting lectures by the course instructor (one each class); on Mondays, that will be followed by a deep dive lecture by another faculty member in PNI, and on Thursdays it will be followed by a student paper presentation.

## **Grading**

Class participation: 50%

Paper presentation: 50%

## **Lectures and Readings**

Note: The readings listed under each lecture are meant to be a resource, for additional background and/or deeper coverage of the material presented in the lectures. Asterisked readings are required.

# Schedule

Week 1 \_\_\_\_\_

## SECTION 1: PERCEPTION, CONSTRAINT SATISFACTION and ASSOCIATIVE LEARNING

### Class 1 (Monday Jan 27) — Overview

- Introduction and History (Cohen)

\***Symbolic vs. Connectionist Models:** Minsky, M. & Papert, S. (1988), *Perceptrons*. Cambridge, MA: MIT Press:

\*Prologue, pp. vii-xv

Epilogue, pp. 247-281

\***Parallel Distributed Processing (PDP).** McClelland, J. and Rumelhart, D. (1986). *Parallel distributed processing: Explorations in the microstructure of cognition. Vol. 1: Foundations*. Cambridge, MA: MIT Press:

\* Chapter 1. The appeal of parallel distributed processing, pp. 3-44

\* Chapter 4. The PDP Framework for Information Processing, pp. 110-137

Chapter 2. A General Framework for Parallel Distributed Processing, pp. 45-76

Chapter 9. An introduction to linear algebra in parallel distributed processing, pp. 365-421

**Spreading Activation Models.** Collins, A. M., & Loftus, E. F. (1975). A spreading-activation theory of semantic processing. *Psychological Review*, 82, 407-428.

**100 Step Challenge.** Feldman, J. A., & Ballard, D. H. (1982). Connectionist models and their properties. *Cognitive Science*, 6(3), 205-254.

**Modularity and Generativity.** Fodor, J. A. & Pylyshyn, Z. W. (1988) Connectionism and cognitive architecture: A critical analysis. *Cognition*, 28, 3-71

**Modeling.** McClelland, J. L. (2009). The place of modeling in cognitive science. *Topics in Cognitive Science*, 1(1), 11-38.

### Class 2 (Thursday Jan 30) — Perception

- Perceptual Phenomena, Constraint Satisfaction and Attractor Networks (Cohen)

- Retina and V1 (Pillow)

[Lab: Necker Cube]

\* **Attractor networks. Hopfield, J. J. (1982).** Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the national academy of sciences*, 79(8), 2554-2558.

\* **The physiology of the retina.** Barlow, HB (1982). The senses, pp 102--113. Cambridge University Press.

\* **Receptive fields of single neurones in the cat's striate cortex.** Hubel, DH & Wiesel, TN (1959). The Journal of Physiology, pp. 148, 574-591.

**Perceptual Bistability.** Gigante G, Mattia M, Braun J, Del Giudice P (2009) Bistable Perception Modeled as Competing Stochastic Integrations at Two Levels. *PLoS Computation Biology* 5(7): e1000430 [<https://doi.org/10.1371/journal.pcbi.1000430>]

**Possible principles underlying the transformation of sensory messages.** Barlow, H (1961). Sensory Communication, Cambridge, MA: MIT Press. pp. 217-234.

**Emergence of simple-cell receptive field properties by learning a sparse code for natural images.** Olshausen BA & Field DJ (1996). *Nature*, 1996, 381, 607-609.

Week 2 \_\_\_\_\_

### Class 3 (Monday Feb 3) — Associative Learning and Topography in Cortex

- Associative Learning and Feature Maps (Cohen)
- Topographic Organization of Visual Cortex (Graziano)

[Lab: Hebbian learning]

\* **Semantic feature organization.** Kohonen, T. (1982). Self-organized formation of topologically correct feature maps. *Biological cybernetics*, 43(1), 59-69.

\* **Cortical Organization of representations.** Graziano, M. S., & Aflalo, T. N. (2007). Mapping behavioral repertoire onto the cortex. *Neuron*, 56(2), 239-251.

**Ocular Dominance Columns.** Miller, K. D., Keller, J. B., & Stryker, M. P. (1989). Ocular dominance column development: Analysis and simulation. *Science*, 245, 605-615.

\* **Hebbian Learning.** Hebb, D. O. (1949). *The Organization of Behavior*. Introduction, xi–xix; and Chapter 4: The first stage of perception: growth of an assembly, 60–78.

**Biological Plausibility.** Song, S., Miller, K. D., & Abbott, L. F. (2000). Competitive Hebbian learning through spike-timing-dependent synaptic plasticity. *Nature neuroscience*, 3(9), 919.

### Class 4 (Thursday Feb 6) — High Level Vision

- Object Recognition and Faces (Gomez)

**Cell-specific object representations.** Desimone, R., Albright, T. D., Gross, C. G., & Bruce, C. (1984). Stimulus-selective properties of inferior temporal neurons in the macaque. *Journal of Neuroscience*, 4(8), 2051-2062.

**“Grandmother cells.”** Gross, C. G. (2002). Genealogy of the “grandmother cell”. *The Neuroscientist*, 8(5), 512-518.

**“Fusiform Face Area” (FFA).** Kanwisher, N., & Yovel, G. (2006). The fusiform face area: a cortical region specialized for the perception of faces. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 361(1476), 2109-2128.

**Distributed object representations.** Haxby, J. V., Hoffman, E. A., & Gobbini, M. I. (2000). The distributed human neural system for face perception. *Trends in cognitive sciences*, 4(6), 223-233.

**Similarity and categorization in vision.** Battleday, R. M., Peterson, J. C., & Griffiths, T. L. (2021). From convolutional neural networks to models of higher-level cognition (and back again). *Annals of the New York Academy of Sciences*, 1505(1), 55-78.

**First connectionist model of neuronal response characteristics.** Andersen, R. A., & Zipser, D. (1988). The role of the posterior parietal cortex in coordinate transformations for visual-motor integration. *Canadian journal of physiology and pharmacology*, 66(4), 488-501.

**Convolutional neural networks.** LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.

**Using deep learning to understand the brain.** Yamins, D. L., & DiCarlo, J. J. (2016). Using goal-driven deep learning models to understand sensory cortex. *Nature Neuroscience*, 19(3), 356-365.

**Circuits in deep learning.** Olah, C., Cammarata, N., Schubert, L., Goh, G., Petrov, M., & Carter, S. (2020). Zoom in: An introduction to circuits. *Distill*, 5(3), e00024-001.

Week 3 \_\_\_\_\_

## SECTION 2: DECISION MAKING - INTEGRATION

### Class 5 (Monday Feb 10) — Dynamics of integration and decision making

- Accumulation to bounds models of Decision Making (Cohen)
- Neural Mechanisms of Perceptual Decision Making (Brody)

[Lab: Drift Diffusion Model]

\* **Integration Models of Decision Making.** Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: a formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), pp.700-720.

\* **Shadlen, M. N., & Newsome, W. T. (2001).** Neural basis of a perceptual decision in the parietal cortex (area LIP) of the rhesus monkey. *Journal of neurophysiology*, 86(4), 1916-1936

**The Drift Diffusion Model.** Ratcliff, R. (1978). A theory of memory retrieval. *Psychological review*, 85(2), 59.

**Brainwide analysis of decision making.** Bondy AG, Charlton, JA, Luo TZ, Kopec CD, Stagnaro WM, Venditto SJC, Lynch L, Janarthanan S, Oline SN, Harris TD & Brody CD (2024). *bioRxiv*, 2024-08.

### Class 6 (Thursday Feb 13) — Optimization of decision making

- Optimization and Control of Decision Making (Cohen)
  - \* **Reward rate optimization of decision making.** Cohen JD & Holmes P (2014). Optimality and some of its discontents. *Topics in Cognitive Science*, 6, 258-278
- Reward rate optimization of decision making.** Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: a formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), pp.720-738.

## SECTION 3: REINFORCEMENT LEARNING - REWARD and NEUROMODULATION

### Class 7 (Monday Feb 17) — Reinforcement Learning

- Reinforcement Learning (Cohen)
- Model-Free vs. Model-Based Learning, Dopamine and Basal Ganglia (Daw)

[Lab: Reinforcement Learning]

- \* **Reward systems.** Schultz, W. (2006). Behavioral theories and the neurophysiology of reward. *Annual Review of Psychology*, 57(1), 87-115.
- \* **Dopamine and reward prediction.** Montague, P. R., Dayan, P., & Sejnowski, T. J. (1996). A framework for mesencephalic dopamine systems based on predictive Hebbian learning. *Journal of Neuroscience*, 16, 1936-1947.
- \* **Model-Free vs. Model-Based RL.** Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature Neuroscience*, 8(12), 1704.

**Model-Free vs. Model-Based RL.** Drummond, N., & Niv, Y. (2020). Model-based decision making and model-free learning. *Current Biology*, 30(15), R860-R865.

**Temporal difference learning.** Sutton, R. S. & Barto, A. G. (1981) Toward a modern theory of adaptive networks: Expectation and prediction. *Psychological Review*, 88, pages 135-170.

**Hierarchical RL.** Botvinick, M. M., Niv, Y., & Barto, A. C. (2009). Hierarchically organized behavior and its neural foundations: a reinforcement learning perspective. *Cognition*, 113(3), 262-280.

### Class 8 (Thursday Feb 20) — Explore/exploit and noradrenergic neuromodulation

- Explore / Exploit and the Locus Coeruleus-Norepinephrine System (Cohen)
- \* **Overview.** Cohen, J. D., McClure, S. M., & Angela, J. Y. (2007). Should I stay or should I go? How the human brain manages the trade-off between exploitation and exploration. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 362(1481), 933-942.
- \* **Adaptive Gain Hypothesis.** Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus-norepinephrine function: adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403-450.

**Behavior and Brain Imaging.** Daw, N. D., O'Doherty, J. P., Dayan, P., Seymour, B., & Dolan, R. J. (2006). Cortical substrates for exploratory decisions in humans. *Nature*, 441(7095), 876.

**Uncertainty and Exploration.** Yu A, & Dayan P (2005). Uncertainty, neuromodulation, and attention. *Neuron*, 46(4), 681-692.

**Dopamine-norepinephrine interactions in exploration and learning.** McClure S, Gilzenrat M and Cohen J (2005). An exploration-exploitation model based on norepinephrine and dopamine activity. *Advances in Neural Information Processing Systems*, 18.

**Biologically-informed model of LC.** Usher M, Cohen JD, Rajkowski J, Kubiak P & Aston-Jones G (1999). The role of locus coeruleus in the regulation of cognitive performance. *Science*, 283, 549-554.

Week 5 —————

## SECTION 4: SEMANTIC MEMORY - STATISTICAL LEARNING and DISTRIBUTED REPRESENTATION

### Class 9 (Monday Feb 24) — Distributed Representation, Learning and Semantics

- Distributed Representations and Semantics (Cohen)
- Backpropagation Learning, Internal Representations and Semantics (Cohen)

[LAB: Rumelhart Network / Backpropagation Learning Algorithm]

- \* **Model of Distributed Memory.** McClelland, J. L., & Rumelhart, D. E. (1985). Distributed memory and the representation of general and specific information. *Journal of Experimental Psychology: General*, 114(2), 159.
- \* **Semantic representations.** Rumelhart, D. E., & Todd, P. M. (1993). Learning and connectionist representations. *Attention and Performance XIV: Synergies in experimental psychology, artificial intelligence, and cognitive neuroscience*, 3-30.
- \* **Semantic representations and coherent covariation.** McClelland, J. L., & Rogers, T. T. (2003). The parallel distributed processing approach to semantic cognition. *Nature Reviews Neuroscience*, 4(4), 310-322.

**Family tree model.** Hinton, G. E. (1986, August). Learning distributed representations of concepts. In *Proceedings of the eighth annual conference of the Cognitive Science Society* (Vol. 1, p. 12).

**Collins, A. M., & Quillian, M. R. (1969). Retrieval time from semantic memory.** *Journal of verbal learning and verbal behavior*, 8(2), 240-247.

**Backpropagation.** Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, 323(6088), 533.

**Nettalk.** Sejnowski, T. J., & Rosenberg, C. R. (1988). NETtalk: A parallel network that learns to read aloud. *Neurocomputing: foundations of research*.

**Dimensional Structure and Learning.** Saxe, A. M., McClelland, J. L., & Ganguli, S. (2019). A mathematical theory of semantic development in deep neural networks. *Proceedings of the National Academy of Sciences*, 116(23), 11537-11546.

**Biological Plausibility.** O'Reilly, R. C. (2001). Generalization in interactive networks: The benefits of inhibitory competition and Hebbian learning. *Neural Computation*, 13(6), 1199-1241.

**Biological Plausibility.** Xie, X., & Seung, H. S. (2003). Equivalence of backpropagation and contrastive Hebbian learning in a layered network. *Neural computation*, 15(2), 441-454.

**AK-to-JDC:** Please verify this is what you meant by "Biological Plausibility / Lazy-Rich learning regimes. Cengiz Pehlevan"

**Biological Plausibility.** Bozkurt, B., Pehlevan, C., & Erdogan, A. (2024). Correlative information maximization: a biologically plausible approach to supervised deep neural networks without weight symmetry. *Advances in Neural Information Processing Systems*, 36.

**Deep Learning Ref** - LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436.

## **Class 10 (Thursday Feb 27) — Bayesian Inference and Neural Networks**

- Bayes in the Age of Intelligent Machines (Griffiths)

- \* **Bayesian models of cognition.** L Griffiths, T., Kemp, C., & B Tenenbaum, J. (2008). *Bayesian models of cognition*.

- \* **Bayesian inference and neural networks.** Griffiths, T. L., Zhu, J. Q., Grant, E., & Thomas McCoy, R. (2024). Bayes in the age of intelligent machines. *Current Directions in Psychological Science*, 33(5), 283-291

- Tenenbaum, J. B., Kemp, C., Griffiths, T. L., & Goodman, N. D. (2011). How to grow a mind: Statistics, structure, and abstraction. *Science*, 331(6022), 1279-1285

Week 6 —————

## **Class 11 (Monday March 3) — Language Processing**

- Word Perception and Verb Learning (Cohen)

- Rules, Generalization, and Language Models (Goldberg)

[No Computational Lab]

- \* **Neurobiology of Language.** Hagoort, P. (2019). The neurobiology of language beyond single-word processing. *Science*, 366(6461), 55-58.

- \* **Past Tense Model.** Rumelhart, D. E. & McClelland, J. L. (1986) On Learning the Past Tenses of English Verbs. PDP, Chapter 18. Rumelhart, D. E., & McClelland, J. L..

- Rules and Exceptions.** Pinker, S. & Prince, A. (1988) On language and connectionism: Analysis of a parallel distributed processing model of language acquisition. *Cognition*, 28, 73-193.

- \* **Letters and Words.** McClelland, J. L., & Rumelhart, D. E. (1981). An interactive activation model of context effects in letter perception: I. An account of basic findings. *Psychological review*, 88(5), 375.

- PDP Model of Word Reading.** Seidenberg, M. S. & McClelland, J. L. (1989) A distributed developmental model of word recognition and naming. *Psychological Review*, 96, 523-568.

## **Class 12 (Thursday March 6) — Brain Systems, Semantics and Context Inference**

- Discourse Processing, Context Inference and Semantics (Cohen)

[Lab: ISC-CI Model] / [LLM Probing exercise]

\* **Deep Learning and Natural Language Processing.** Hirschberg, J., & Manning, C. D. (2015). Advances in natural language processing. *Science*, 349(6245), 261-266.

**Syntax in LLMs.** Hewitt, J., & Manning, C. D. (2019, June). A structural probe for finding syntax in word representations. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1* (Long and Short Papers) (pp. 4129-4138).

**Latent Semantic Analysis.** Dumais, S. T. (2004). Latent semantic analysis. *Annual Review of Information Science and Technology (ARIST)*, 38, 189-230.

**Word2Vec:** Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, 26.

**BERT.** Tenney, I, Das D & Pavlick E (2019). BERT rediscovers the classical NLP pipeline. *arXiv preprint arXiv:1905.05950*.

## Spring Break

Week 7 —————

### SECTION 5: EPISODIC MEMORY - BINDING

Class 13 (Monday March 17) — Complementary learning systems and hippocampal function / oscillatory inhibition, sleep and PTSD (**Norman**)

#### [LAB: EGO Model]

\* **Episodic memory.** Clewett D., DuBrow S., & Davachi L. (2019). Transcending time in the brain: How event memories are constructed from experience. *Hippocampus*, 29:162-183.

**Episodic memory.** Tulving, E. (2002). Episodic memory: From mind to brain. *Annual review of Psychology*, 53(1), 1-25.

\* **Complementary learning systems.** McClelland J.L., McNaughton B.L., & O'Reilly R.C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *The Psychological Review*, 102:419-457.

**Complementary learning systems (extended).** O'Reilly R.C., & Norman K.A. (2002). Hippocampal and neocortical contributions to memory: Advances in the complementary learning systems framework. *Trends in Cognitive Sciences*, 6(12):505-510.

**Temporal Context Model.** Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of mathematical psychology*, 46(3), 269-299.

**CLS theory updated.** Kumaran, D., Hassabis, D., & McClelland, J. L. (2016). What learning systems do intelligent agents need? Complementary learning systems theory updated. *Trends in cognitive sciences*, 20(7), 512-534.

**Latent cause inference.** Gershman, S. J., & Niv, Y. (2010). Learning latent structure: carving nature at its joints. *Current opinion in neurobiology*, 20(2), 251-256.



AK-to-JDC: Please verify this is what you meant by "Oscillatory inhibition"

**Oscillatory inhibition.** Norman, K. A., Newman, E. L., & Detre, G. (2007). A neural network model of retrieval-induced forgetting. *Psychological review*, 114(4), 887.

AK-to-JDC: I can't find the "BCM" you mentioned.

## SECTION 6: ATTENTION, WORKING MEMORY AND COGNITIVE CONTROL - MODULATION

Class 14 (Thursday March 20) — Selective attention, automaticity and control

\* **Empirical Study of Automaticity, Attention and Control.** Shiffrin, RM & Schneider W (1977). Controlled and automatic human information processing: II. Perceptual learning, automatic attending and a general theory. *Psychological review*, 84(2), 127.

\* **Prefrontal Cortex and Control.** Miller EK & Cohen JD (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24, 167-202.

**Forms of Attention.** Chun, M. M., Golomb, J. D., & Turk-Browne, N. B. (2011). A taxonomy of external and internal attention. *Annual Review of Psychology*, 62(1), 73-101.

**Connectionist Model of Attention, Automaticity and Control.** Cohen JD, Dunbar K & McClelland JL (1990). On the control of automatic processes: A parallel distributed processing model of the Stroop effect. *Psychological Review*, 97(3), 332-361.

Week 8 \_\_\_\_\_

Class 15 (Monday March 24) — Working memory, oscillatory dynamics and prefrontal function (**Buschman**)

[LAB: Stroop Model]

TODO: Bouchacourt et al. (is the paper below what you had in mind?)

Bouchacourt, F., Tafazoli, S., Mattar, M. G., Buschman, T. J., & Daw, N. D. (2022). Fast rule switching and slow rule updating in a perceptual categorization task. *Elife*, 11, e82531.

TODO: Fusi

Panichello, M. F., DePasquale, B., Pillow, J. W., & Buschman, T. J. (2019). Error-correcting dynamics in visual working memory. *Nature communications*, 10(1), 3366.

Class 16 (Thursday March 27) — Shared vs. separated representations, compositional vs. conjunctive coding & capacity constraints; Miller's Law  
[Multitasking network]

TODO: JDC

\_\_\_\_\_

## Week 9

### Class 17 (Monday March 31) — Higher Level Cognition: EM-WM interactions and Abstraction [ART tasks / ESNB]

- \* **Attention and Binding.** Treisman, A. (1998). Feature binding, attention and object perception. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 353(1373), 1295-1306.

**Episodic vs. Working Memory.** Beukers AO, Hamin M, Norman KA & Cohen JD (2024). When working memory may be just working, not memory. *Psychological Review*, 131(2), 563. <https://psyarxiv.com/jtw5p>

**Semantics, Context and Control.** Giallanza T, Campbell D, Cohen JD & Rogers TT (2024). An integrated model of semantics and control. *Psychological Review*. <https://psyarxiv.com/jq7ta>.

**Semantics, Control and Context Inference.** Giallanza T, Rogers TT & Cohen JD. (under review). An integrated model of semantics and control, Part 2: Solving the similarity paradox through context inference. <https://osf.io/preprints/psyarxiv/fxc87>.

**External memory in neural networks: The Neural Turing Machine.** Graves, A. (2014). Neural Turing Machines. arXiv preprint arXiv:1410.5401.

**Episodic Memory and Abstraction.** Webb TW, Sinha I & Cohen JD (2021). Emergent symbols through binding in external memory. *ICLR 2021: Proceedings of the International Conference on Learning Representations*. <https://arxiv.org/abs/2012.14601>

#### **TODO: Relational Bottleneck: TICS**

- \* **Episodic Generalization and Control.** Giallanza T, Campbell D & Cohen JD (2024). Toward the emergence of intelligent control: Episodic generalization and optimization. *OpenMind*, 8, 688-722. <https://osf.io/preprints/psyarxiv/dzvpv/>
- \* **Connectionist vs. Symbolic Processing.** Fodor, J. A., & Pylyshyn, Z. W. (1988). Connectionism and cognitive architecture: A critical analysis. *Cognition*, 28(1-2), 3-71.

**Connectionist Model of Symbol Processing.** Touretzky, D. S., & Hinton, G. E. (1985). Symbols among the neurons: Details of a connectionist inference architecture. In *IJCAI* (Vol. 85, pp. 238-243).

**Analogical Reasoning and LLMs.** Webb, T., Holyoak, K.J. & Lu, H. Emergent analogical reasoning in large language models. *Nat Hum Behav* 7, 1526–1541 (2023). <https://doi.org/10.1038/s41562-023-01659-w>

**In-context Learning and Induction Heads.** Olsson, C., Elhage, N., Nanda, N., Joseph, N., DasSarma, N., Henighan, T., ... & Olah, C. (2022). In-context learning and induction heads. arXiv preprint arXiv:2209.11895.

**Tensor Product Representations.** Plate, T. A. (1995). Holographic reduced representations. *IEEE Transactions on Neural networks*, 6(3), 623-641.

**Tensor Product Representations and Symbolic Processing.** Smolensky, P. (1990). Tensor product variable binding and the representation of symbolic structures in connectionist systems. *Artificial intelligence*, 46(1-2), 159-216.

**Tensor product representations and Deep Learning.** McCoy, R. T., Linzen, T., Dunbar, E., & Smolensky, P. (2018). RNNs implicitly implement tensor product representations. arXiv preprint arXiv:1812.08718.

**LLM content effects.** Dasgupta, I., Lampinen, A. K., Chan, S. C., Creswell, A., Kumaran, D., McClelland, J. L., & Hill, F. (2022). Language models show human-like content effects on reasoning. arXiv preprint arXiv:2207.07051, 2(3).

## Class 18 (Thursday April 3) — Optimization and control - EVC

- \* **Conflict Monitoring.** Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological review*, 108(3), 624.
- \* **Expected Value of Control.** Shenhav A, Botvinick MM & Cohen JD (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron*, 79, 217-240.
- \* **Resource rational analysis.** Lieder, F., & Griffiths, T. L. (2020). Resource-rational analysis: Understanding human cognition as the optimal use of limited computational resources. *Behavioral and brain sciences*, 43, e1.

**Bounded Rationality.** Howes, A., Lewis, R. L., & Vera, A. (2009). Rational adaptation under task and processing constraints: implications for testing theories of cognition and action. *Psychological review*, 116(4), 717.

**Resource Rationality and Effort.** Shenhav A, Musslick S, Lieder F, Kool W, Griffiths TL, Cohen JD & Botvinick MM (2017). Toward a rational and mechanistic account of mental effort. *Annual Review of Neuroscience*, 40, 99-124.

Week 10 \_\_\_\_\_

## SECTION 7: MOTOR FUNCTION

### Class 19 (Monday April 7) — Motor representations, compositionality, and dynamics (Graziano / Buschman)

[LAB: Sequential adjustment effects / EVC model]

### Class 20 (Thursday April 10) — Motor control, learning and cerebellar function (Taylor)

Week 11 \_\_\_\_\_

## SECTION 8: DEVELOPMENT and SOCIAL COGNITION

### Class 21 (Monday April 14) — Development

**Infantile amnesia.** Howe, M. L., & Courage, M. L. (1993). On resolving the enigma of infantile amnesia. *Psychological bulletin*, 113(2), 305.

Baillargeon, R. (1987). Object permanence in 3½- and 4½-month-old infants. *Developmental psychology*, 23(5), 655.

**Infantile Amnesia.** Ellis, C. T., Skalaban, L. J., Yates, T. S., Bejjanki, V. R., Córdova, N. I., & Turk-Browne, N. B. (2021). Evidence of hippocampal learning in human infants. *Current Biology*, 31(15), 3358-3364.

## Class 22 (Thursday April 17) — Social cognition (**Hasson**)

**Neural coupling.** Hasson, U., Ghazanfar, A. A., Galantucci, B., Garrod, S., & Keysers, C. (2012). Brain-to-brain coupling: a mechanism for creating and sharing a social world. *Trends in cognitive sciences*, 16(2), 114-121.

AK-to-JDC: Please verify this is what you meant by "Dynamic programming models"

**Bayesian model of pragmatics.** Goodman, N. D., & Frank, M. C. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in cognitive sciences*, 20(11), 818-829.

AK-to-JDC: Please verify this is what you meant by "Theory of Mind - Baron-Cohen"

**Theory of mind and autism.** Baron-Cohen, S., Leslie, A. M., & Frith, U. (1985). Does the autistic child have a "theory of mind"? *Cognition*, 21(1), 37-46

**Brain regions responsible for social cognition.** Saxe, R. (2006). Uniquely human social cognition. *Current opinion in neurobiology*, 16(2), 235-239.

Week 12

---

## SECTION 9: DISORDERS

### Class 23 (Monday April 21) — Cognitive Neuropsychology (**Kastner**) / Origins of Computational Psychiatry

Cohen, J. D., & Servan-Schreiber, D. (1992). Context, cortex, and dopamine: a connectionist approach to behavior and biology in schizophrenia. *Psychological review*, 99(1), 45.

### Class 24 (Thursday April 24) — Modern Computational Psychiatry (Cohen/**Niv**)

Pisupati, S., Langdon, A. J., Konova, A. B., & Niv, Y. (2024). The utility of a latent-cause framework for understanding addiction phenomena. *Addiction Neuroscience*, 10, 100143.

AK-to-JDC: Please verify this is what you meant by "Hartley & Daw - big data"

Decker, J. H., Otto, A. R., Daw, N. D., & Hartley, C. A. (2016). From creatures of habit to goal-directed learners: Tracking the developmental emergence of model-based reinforcement learning. *Psychological science*, 27(6), 848-858.